

Dart Tutorial 7: Lists in Dart

This chapter explains how to store and manage collections of data in Dart using Lists. After completing this tutorial, students will understand how to create, access, and modify Lists using built in methods.

1. Introduction to Lists

A **List** in Dart is an ordered collection of items, similar to an array in other programming languages.

Lists can store multiple values of the same type, such as numbers, strings, or objects, in a single variable.

2. Why We Use Lists in Dart

In real world apps, we often need to store and manage multiple related values, such as a list of products or usernames.

Lists allow us to:

- Store multiple values in a single variable.
 - Access and update items easily.
 - Loop through items efficiently.
 - Add or remove items dynamically.
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3. Creating a List

```
List<String> fruits = ['Apple', 'Banana', 'Mango'];  
print(fruits);
```

Output:

```
[Apple, Banana, Mango]
```

In this example, `List<String>` means this list can only store String values.

4. Accessing List Items

List items are accessed using an **index**, which starts from 0.

```
List<String> fruits = ['Apple', 'Banana', 'Mango'];  
  
print(fruits[0]);  
print(fruits[2]);
```

Output:

```
Apple  
Mango
```

5. Checking List Properties

Property	Description	Example
length	Number of items	fruits.length → 3
isEmpty	Checks if list has no items	fruits.isEmpty → false
isNotEmpty	Checks if list has items	fruits.isNotEmpty → true
first	Returns first item	fruits.first → Apple
last	Returns last item	fruits.last → Mango

6. Adding Items to a List

```
List<String> fruits = ['Apple', 'Banana'];  
  
fruits.add('Mango');  
print(fruits);
```

Output:

```
[Apple, Banana, Mango]
```

Adding multiple items

```
fruits.addAll(['Orange', 'Grapes']);  
print(fruits);
```

7. Removing Items from a List

```
List<String> fruits = ['Apple', 'Banana', 'Mango'];  
  
fruits.remove('Banana');  
print(fruits);
```

Output:

```
[Apple, Mango]
```

Removing by index

```
fruits.removeAt(0);  
print(fruits);
```

8. Updating List Items

```
List<String> fruits = ['Apple', 'Banana', 'Mango'];  
  
fruits[1] = 'Grapes';  
print(fruits);
```

Output:

```
[Apple, Grapes, Mango]
```

9. Looping Through a List

```
List<String> fruits = ['Apple', 'Banana', 'Mango'];  
  
for (String fruit in fruits) {
```

```
print(fruit);  
}
```

Output:

```
Apple  
Banana  
Mango
```

10. Useful List Methods

Method	Description	Example
sort()	Sorts the list	fruits.sort()
reversed	Reverses list order	fruits.reversed
contains()	Checks if item exists	fruits.contains('Apple')
indexOf()	Finds index of an item	fruits.indexOf('Mango')
clear()	Removes all items	fruits.clear()

```
List<int> numbers = [5, 3, 8, 1];  
numbers.sort();  
print(numbers);
```

Output:

```
[1, 3, 5, 8]
```

11. Conclusion of Tutorial 7

In this tutorial, students have learned:

1. What Lists are and why they are used.
2. How to create and access a List.
3. How to add, remove, and update List items.
4. How to loop through a List.
5. How to use common List methods like sort, contains, and reversed.

In the next tutorial, we will learn about Sets and Maps in Dart.

This completes Dart Tutorial 7.